

Lecture 6

September 10

Probability...

Who is De Moivre and how much is enough?
(Nic)

Learning objectives

- Calculate the mean, variance, and standard deviation of Bernoulli and binomial random variables, and interpret the units of each quantity.
- Explain why expectations add even for dependent variables, while adding variances requires additional assumptions; compare independent coin tosses with counting the same toss twice.
- Explain why binomial counts fluctuate on a scale proportional to $\sqrt{n}$, while sample proportions fluctuate on a scale proportional to $1/\sqrt{n}$ for fixed p .
- Describe how de Moivre approximated the symmetric binomial distribution with a smooth curve and used areas under that curve to approximate probabilities.

Axioms of Probability

Non-negativity: if E is an event $P(E) \geq 0$

Normalization: if S is the event that anything can happen $P(S) = 1$

Additivity: if E_1 and E_2 are independent events then $P(E_1 \cup E_2) = P(E_1) + P(E_2)$

Bernoulli random variables

$$y \sim \text{Bernoulli}(p)$$

" $\sim$ " means distributed as

$$P(y = \textcolor{blue}{1}) = p$$

$\textcolor{red}{H}$
100

$$P(y = \textcolor{blue}{0}) = 1 - p$$

$\textcolor{red}{T}$
75

$P(y=y)$ probab. mass function

when we are working with $y = \begin{cases} 0 & \text{prob } 1-p \\ 1 & \text{prob } p \end{cases}$

$$P(y=y) = p^y (1-p)^{1-y}$$

Expectations

The mean of a random variable

$$E[Y] = \sum_y y P(Y=y)$$

$$Y = \begin{cases} 0 & 1-p \\ 1 & p \end{cases}$$

$$E[Y] = 0 \cdot (1-p) + 1 \cdot p = p$$

Binomial

Expected value of a Binomial

$$X \sim \text{Binom}(n, p)$$

$$\begin{aligned} E[X] &= \sum_{k=0}^n k \cdot P(X=k) = 0 \cdot P(X=0) + 1 \cdot P(X=1) + \dots + n \cdot P(X=n) \\ &= 0 + \binom{n}{1} p (1-p)^{n-1} + 2 \binom{n}{2} p^2 (1-p)^{n-2} \\ &\quad + \dots + n \binom{n}{n} p^n (1-p)^0 \end{aligned}$$

$$\begin{aligned} P(X=k) &= \binom{n}{k} p^k (1-p)^{n-k} \\ &= \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \end{aligned}$$

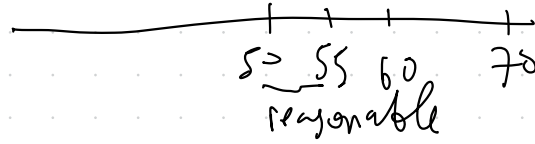
$$\begin{aligned} E[X] &= E[Y_1 + Y_2 + \dots + Y_n] = E[Y_1] + E[Y_2] + \dots + E[Y_n] \\ &\quad \left. \begin{array}{l} Y_1 \sim \text{Bern}(p) \\ Y_2 \sim \text{Bern}(p) \\ \vdots \\ Y_n \sim \text{Bern}(p) \end{array} \right\} \begin{array}{l} \text{independent} \\ \text{of each} \\ \text{other} \end{array} \end{aligned}$$

Linearity of expectations

$$= \overbrace{p + p + \dots + p}^{n \text{ times}} = np$$

If $X \sim \text{Binomial}(n, p)$, how much does it vary around np ?

$$n = 100, p = \frac{1}{2}$$



Variance: fluctuation of a random variable about its mean/expectation in squared units

$$\begin{aligned}\text{Var}(X) &= E[(X - EX)^2] \\ &= E[X^2 - 2XE(X) + (E(X))^2] \\ &= E[X^2] + E[-2XE(X)] + E[(E(X))^2] \\ &= \underline{E[X^2]} - 2E(X)E(X) + (E(X))^2 E[1] \\ &= E[X^2] - 2(E(X))^2 + (E(X))^2 = E[X^2] - E(X)^2\end{aligned}$$

Variance of a Bernoulli:

$$Y \sim \text{Bern}(p)$$

$$E[Y] = p$$

$$E[Y^2] = ?$$

$$= E[Y] = p$$

$$Y^2 = Y$$

$$Y=1, Y^2=1$$

$$Y=0, Y^2=0$$

$$\text{Var}(Y) = E[Y^2] - E[Y]^2 = p - p^2 = p(1-p)$$

$$\begin{matrix} 0 & 1 \\ -p & 1-p \end{matrix}$$

Standard deviation is $SD(Y) = \sqrt{\text{Var}(Y)}$

$$\text{For Bern } SD(Y) = \sqrt{p(1-p)}$$

$$\text{Var}(Y) = E[(Y - E(Y))^2] = \sum \dots$$

$$= \underbrace{-p \text{ w.p. } 1-p}_{\text{}} + \underbrace{(1-p) \text{ w.p. } p}_{\text{}}$$

Variance of a Binomial

$$\text{Var}(X) = np(1-p)$$

$$\text{SD}(X) = \sqrt{np(1-p)}$$

Next class

Is Variance linear?

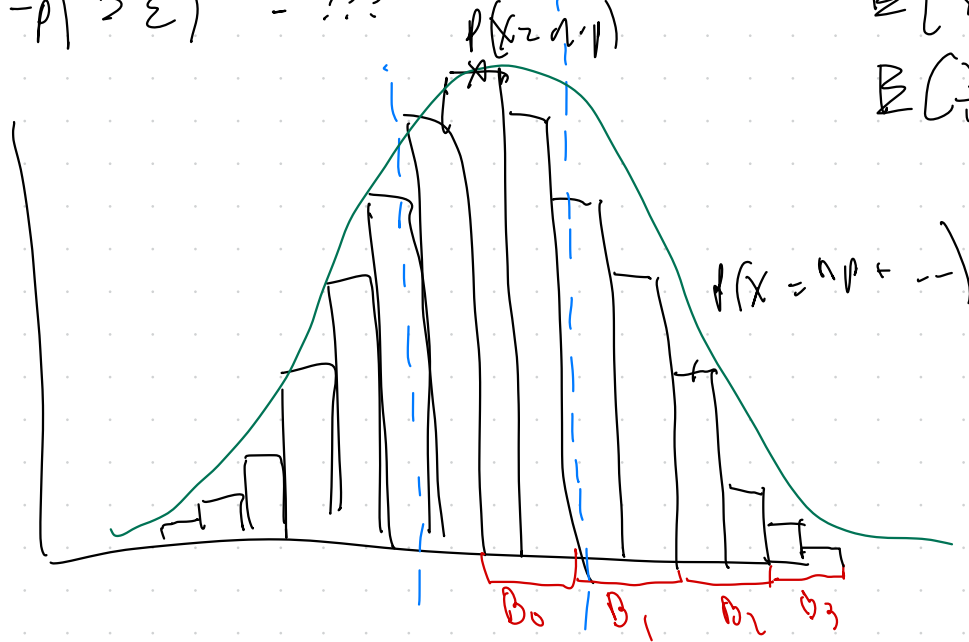
Next
Class...

Bach & Bernoulli and De Moivre

$$P(|X/n - p| > \varepsilon) = ???$$

$$\mathbb{E}[x] = np$$

$$\mathbb{E}\left[\frac{x}{n}\right] = \frac{np}{n} = p$$



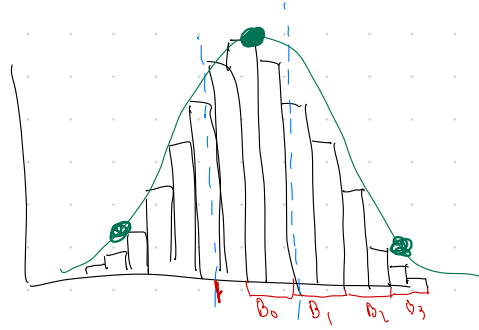
Goal:

De Moivre's approach

For $X \sim \text{Binomial}(n, \frac{1}{2})$

$P(X = n/2 + l) \approx P(X = n/2) \exp\left(-\frac{l^2}{n/4}\right)$
- a lot

$$\approx \underbrace{\frac{2}{\sqrt{n \cdot 2\pi}} \exp\left(-\frac{l^2}{n/4}\right)}$$



further, writing l for p and noting “that if m or $\frac{1}{2}n$ be a Quantity infinitely great, then the Logarithm of the Ratio, which a Term distant from the middle by the Interval l , has to the middle Term, is $-2ll/n$ ” (De Moivre, 1738, p. 237; 1756, p. 245; see Figure 2.2). What De Moivre had shown

Logarithms

SHERWIN'S Mathematical Tables,

Contriv'd after a most comprehensive METHOD:

CONTAINING,

Dr. *Wallis's* Account of Logarithms, Dr. *Halley's* and Mr. *Sharp's* Ways of constructing them; with Dr. *Newton's* contraction of *Briggs's* Logarithms,

VIZ.

A TABLE of Logarithms of the Numbers from 1 to 101000, with the means to find readily the Logarithm of any Number, and the Number of any Logarithm, to seven places of Figures:

AND

TABLES of natural and logarithmic Sines, Tangents, Secants, and Verfed-sines, to every minute of the Quadrant:

WITH THE
EXPLICATION and USE prefix'd.

The third EDITION.

Carefully revised and corrected,

By *WILLIAM GARDINER*.

L O N D O N :

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Tablas prima.

Num. abito	Logarithmi.	Num. abito	Logarithmi.	Num. abito	Logarithmi.
1	0,00000,00000	34	1,51147,32179,412	67	1,51607,48017,0081
2	0,30103,00000	35	1,51147,32179,412	68	1,51607,48017,0081
3	0,47712,12547,1966	36	1,51147,32179,412	69	1,51607,48017,0081
4	0,60206,00000	37	1,51147,32179,412	70	1,51607,48017,0081
5	0,69897,00000	38	1,51147,32179,412	71	1,51607,48017,0081
6	0,77815,1401,8164	39	1,51147,32179,412	72	1,51607,48017,0081
7	0,84509,00000	40	1,51147,32179,412	73	1,51607,48017,0081
8	0,89010,00000	41	1,51147,32179,412	74	1,51607,48017,0081
9	0,95424,14004,1912	42	1,51147,32179,412	75	1,51607,48017,0081
10	1,00000,00000	43	1,51147,32179,412	76	1,51607,48017,0081
11	1,04139,10881,581	44	1,51147,32179,412	77	1,51607,48017,0081
12	1,07918,14000,4752	45	1,51147,32179,412	78	1,51607,48017,0081
13	1,11394,15121,0034	46	1,51147,32179,412	79	1,51607,48017,0081
14	1,14618,16136,7844	47	1,51147,32179,412	80	1,51607,48017,0081
15	1,17609,17190,5560	48	1,51147,32179,412	81	1,51607,48017,0081
16	1,20411,18281,5591	49	1,51147,32179,412	82	1,51607,48017,0081
17	1,23048,19411,7817	50	1,51147,32179,412	83	1,51607,48017,0081
18	1,25527,20591,0311	51	1,51147,32179,412	84	1,51607,48017,0081
19	1,27871,21821,2812	52	1,51147,32179,412	85	1,51607,48017,0081
20	1,30099,23101,5313	53	1,51147,32179,412	86	1,51607,48017,0081
21	1,32211,24421,7814	54	1,51147,32179,412	87	1,51607,48017,0081
22	1,34211,25781,1015	55	1,51147,32179,412	88	1,51607,48017,0081
23	1,36101,27191,3516	56	1,51147,32179,412	89	1,51607,48017,0081
24	1,37871,28651,6017	57	1,51147,32179,412	90	1,51607,48017,0081
25	1,39531,30161,8518	58	1,51147,32179,412	91	1,51607,48017,0081
26	1,41081,31721,1019	59	1,51147,32179,412	92	1,51607,48017,0081
27	1,42531,33331,3520	60	1,51147,32179,412	93	1,51607,48017,0081
28	1,43881,34991,6021	61	1,51147,32179,412	94	1,51607,48017,0081
29	1,45141,36701,8522	62	1,51147,32179,412	95	1,51607,48017,0081
30	1,46311,38461,1023	63	1,51147,32179,412	96	1,51607,48017,0081
31	1,47401,40271,3524	64	1,51147,32179,412	97	1,51607,48017,0081
32	1,48411,42131,6025	65	1,51147,32179,412	98	1,51607,48017,0081
33	1,49341,44041,8526	66	1,51147,32179,412	99	1,51607,48017,0081
34	1,50201,46001,1027	100	1,51607,48017,0081		

De Moivre's second attempt

$$\text{write } e(-x) = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots$$

$$\int e^{-x} dx = \int (\quad \quad \quad) dx$$

$$= \int 1 dx - \int x dx + \int \frac{x^2}{2!} dx - \dots$$

Table 2.1. Values of the normal integral $\int_{-a}^a (1/\sqrt{2\pi})e^{-x^2/2}dx$, computed by De Moivre in 1733 as approximations to binomial probabilities $P(\frac{1}{2}n - \frac{1}{2}a\sqrt{n} \leq X \leq \frac{1}{2}n + \frac{1}{2}a\sqrt{n})$ and computed exactly by modern methods.

a	De Moivre	Exact
1	0.682688	0.682689
2	0.95428	0.95450
3	0.99874	0.99730

Sources: De Moivre (1733, pp. 4–6; 1738, pp. 238–241; 1756, pp. 246–249) and modern tables.

Note: As this table shows, De Moivre's calculations were quite accurate. Surprisingly, though, he tripped up at one point in converting these probabilities to odds. In the original 1733 pamphlet he gave the odds of exceeding the given limits correctly, as, respectively, 28 to 13, 21 to 1, and 792 to 1. In later publications, however, (for example, De Moivre, 1756, pp. 248, 251) this last odds ratio appeared as 369 to 1. Apparently De Moivre had mistakenly convinced himself that he had forgotten to double the probability 0.00126 of exceeding $(3/2)\sqrt{n}$, and he doubled it to get 0.00252 or odds of 396 to 1, then reversed the last two digits giving 369 to 1.

Are we happy?

Bern.

$$N = 25500$$

→ be within $\frac{1}{50}$ of $\frac{30}{30}$
with 1000:1 odds

De Moivre

$$N = 6500$$

2 The proof of Law of Large Numbers.

Chebyshev's Inequality, $P(|\frac{X}{n} - p| > \varepsilon) \leq \frac{\text{Var}(\frac{X}{n})}{\varepsilon^2}$

for any $\varepsilon > 0$.

Goal $P(|\frac{X}{n} - p| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$ for any fixed $\varepsilon > 0$

If $X \sim \text{Binomial}(n, p)$,

$\text{Var}(X)$

$\text{Var}(X/n)$